ZOO501 FINAL TERM MCQS

1. Mode of commitment_____3___
2. strageries of commitment_____2___
3. Rana pipiens usually lays around __2500 eggs____while the bullfrog, Rana catesbiana, can lay as many as 20,000
4. Multinucleated type of cell form._____syncytiotrophoblast
5. slow block, initiated by the intracellular release of_____ calcium ions
6. There are_____ two ____ major modes of bone formation, or **osteogenesis**
7. Mutations of the ___*SOX9* gene____cause camptomelic dysplasia, a rare disorder of skeletal development that results in deformities of most of the bones of the body
8. Human brain contains_____**10^{12}** glial cells____
9. Telolecithal is cleavage is found in eggs of _____Birds and Fishes
10. ____Pax6 _____ plays a major role in eye formation
11. ___Pax1_____and Scleraxis are the transcription factors are thought to activate cartilage-specific genes
12. The eggs of insects have their yolk in the ____center_____ (i.e., they are centrolecithal),
13. Egg activates due to which ion more _____Ca____
14. Which dermis give rise to brain _____Ectoderm____
15. Which dermis give rise to eye _____Ectoderm_____
16. Mod of commitment in insects _____syncytial specification____
17. Regulative development arises from_____ conditional specification
18. Nucleosome is composed of an octamer of histone proteins wrapped with two loops containing approximately ___140 base pairs___ of DNA
19. the membrane potential of the sperm becomes ___more negative___ as potassium ions leave the sperm
20. Paracrine factors such as TGF- and FG**F7** are important in normal _____skin development
21. The male leopard ___frog___ release sperm in the summer
22. In mouse ZP3 is attached to_____zona pellucid
23. It is estimate ____egg is 25000 to 50000 mRNA. _____Sea Urchin
24. Fertilin protein is important for sperm-egg fusion in _____Mammals___
25. Fast block to polyspermy is initiated by _____Sodium___
26. Transition from fertilization to cleavage is caused by factor _____MPF
27. Eggs having sparse and equally yolk space are called _____isolethial___
28. Archenteron is used for _____primitive gut___
29. Spiral cleavage found in _____Snails____
30. Sonic hedgehog is a type of _____protein____
31. Which type of neurulation in fish _____secondary___
32. Nural crest divides functionally in domains _____4_____
33. Myogenic bHLH proteins are specific for development of _____muscle cell___

34. Zygote which divide mitotically to produce all the cells of the body.
35. Tatratology the study of birth defect.
36. Identical twins occur in roughly 0.25% of human births.
37. Requlatively development is seen in most vertebrate embroys , and it is obviously critical in development of identical twins.
38. Insect such as drosophila use all three mods of specification to commit their cells to particular fates.
39. The speed of metamorphosis is directly linked with environment pressures
40. In acetabularia unicellular organisms, development us controlled at both the transciptional and translational level.
41. The individuals of each species, however is divided into two mating types Plus and minus calamadomonas.
42. 140 base pairs of DNA nucleosome.
43. Viteline envelop contains at least eight different glycoproteins.
44. The first evidence suggestion the importance of sperm in reproduction came from a series experiment performed by Lazzaro Spallanzani
45. The volume of the sea urchin is about 200 Pico liters
46. Hindbrain Rhombencephalon
47. The most crucial, difference between mammalian cleavage and all other types involves the phenomenon of compaction
48. The embryo contains three germ layers. The ectoderm, the endoderm and the mesoderm.
49. The neural form keel fish
50. Spiral holoblastic clevage is characteristic of several animals groups, including annelid worms, some flateworms, and most mollusks(snails).
51. Diploblastic animals are those who have ectoderm and endoderm but no true mesoderm (hydra)
52. Cell fate is determined by signalling. The micromeres constitute a major signalling center. B-catenin is important for the inducing capacity of micromeres.
53. Even distribution of yolk in ovum isolecithal
54. Size of bindin protein 30,500-Da
55. which has two parts in cell cycle options Blastomere
56. After the blastoderm cells have covered about half the zebra fish yolk cell, a thickening occurs throughout the margin of the epibolizing blastoderm. This thickening is called the **germ ring**
57. Which forms spinal cord, hind brain and mid brain chordamesoderm.
58. This streak is first visible in birds as a thickening of the Epiblast at the posterior region of the embryo, just anterior to koller sickle.
59. I will induce the uterine cells of the mammals to from the matrenal portion of the placenta, the Decidua.
60. Between the area pellucida and the area opaca is a thin layer of cells called the **marginal zone** (or marginal belt).
61. The mitotic division of the nucleus is called karyokinesis
62. Fishes, reptiles , and birds undergo discoidal meroblastic cleavage, wherein the early cell divisions do not cut through the yolk of the egg.
63. The fast block, initiated by sodium influx into the cell in sea urchin.

64. The cells adjacent to the chordamesoderm, the paraxial mesoderm cells, are the precursors of the mesodermal somites in the fish.

65. The rudiment of the central nervous system, is called neurulation.

66. Blastula produced by radial cleavage has no blastocoel and are called stereoblastulae.

67. Fraternal twins are the result of two separate fertilization events.

68. The prechordal ____plate_____induces the formation of the forebrain; the chordamesoderm induces the formation of the midbrain, hindbrain, and spinal cord.

69. The resulting larvae of snail is called _____glochidia____which resemble tiny bear traps.

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1. Due to **Loss** in hydrogen ions, pH rises.
2. As a result of gastrulation, ectoderm, mesoderm and **Endoderm** are formed.
3. Male leopard **Frogs** make their sperm in summer.
4. **Differential gene transcription** is step for gene expression regulation.
5. Coding regions of DNA **exon.**
6. How many domains are present for transcription **3**
7. pdx1 is present in **Pancreatic** And Duodenal1 gene.
8. Bacteria transmit genes from one cell to other by **sex pili.**
9. Cells rearrange their **DNA** to create new immunoglobin and antigen receptor gene.
10. Sea urchins contain **25,000 to 50,000** mRNA.
11. In many species, sperm are attracted towards eggs by **chemotaxis**.
12. Acrosomal protein mediating recognition by protein **bindin.**
13. Sea water has **High** sodium ion conc.
14. Eggs have sparse **equally** spaced yolk are icolecithel.
15. Cytoplasm of blastodisc become embryo this meroblastic division called **discoidal.**
16. **The micromere** constitute major signaling center.
17. Osteoblasts secrete collagen proteoglycan **matrix** that is able to bind calcium salts
18. Tgf beta family secreted from the surface **ectoderm** and roof of neural tube.
19. Dendrites are numerous and do not extend far from the neuronal cell body or **soma,** axons may extend for several feet.
20. One agent of ventral specification is **sonic** hedgegog protein.
21. In frogs and chicks, **secondary** nueralation is seen in neural tube.
22. The cells at dorsal most portion of neural tube become **neural crest cells.**
23. About **33%** of identical twins have two complete and separate chorions
24. Mammals undergo holoblastic **rotational** cleavage.
25. In birds, **gravity** is critical in determining anterior posterior axis
26. Decidua becomes rich in blood vessels that will provide **oxygen** and nutrients to embryo.
27. Mammalian oocyte is released from ovary and swept by fimbriae into **the oviduct.**
28. Fertilization in chick egg occurs in **oviduct.**
29. Developmental decisions are usually **irreversible.**
30. All multicellular organisms arise by a slow process of progressive change called **development**.
31. Major class of cadherin includes **N- cadherin.**
32. Only **small %** of genome is expressed in partial cell.
33. Cortical are **granules** are membrane bound structure containing proteolytic enzymes.
34. Small subunits of MPF **cyclin-dependent kinase.**
35. It secretes a **cellulose** tunic (which gave the name "tunicates" to these creatures).
36. Inner surface cells thin 30×5um process **filopodia.**
37. β-catenin is Important for Inducing capacity of **micromeres.**
38. In fishes , First **12** divisions occur synchronously form a mouln of cells that sits an animal pole of large yolk cell.
39. Fertilization occurs in **ampulla** of oviduct a close, a region close to ovary.
40. In amphibians most of the xenopus most of tadpole is formed by **primary neurulation** and tail of neural tube formed by 2ndry neuralation.
41. Neural fold migrates to midline of embryo and fusing to form **neural tube** beneath the overlying ectoderm.

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42. **Tissues** form nervous system in the fly or in the frog.
43. The marginal zones to form clusters of neurons called **nuclei.**
44. Paracrine factors instruct **Myotome cells** to become muscles by inducing them to synthesize MyoD protein.
45. Sclerix is expressed in mesenchyme **from** the scleretome.
46. Acrosomal reaction in **sea urchins** is often highly specific to each species.
47. Cells are arrange their.. to creat new immunoglobin
48. In frogs and chicks, **secondary neurulation** is usually seen in the neural tube of the lumbar (abdominal) and tail vertebrae.
49. Regulation of gene expression by **several level.**
50. Which type of neuralation in fish **exclusively secondary**.
51. sonic hedgehocy is a type of **protein.**
52. **Blastomeres** have two parts of cell cycle.
53. Number of cortical granules Sea urchin **15000.**
54. How many strategies involve in cell commitment **2.**
55. Resting membrane potential of the egg is **70mv.**
56. One chemotactic molecule, a 14-amino acid peptide called **resact**.
57. Force for sperm propulsion is provided by **dynin.**
58. Regulative development arises from **Conditional Specification.**
59. Human brain contains **10^{12} glial cells.**
60. Fish have discoidal **telolicithal** pattern.
61. Process of osteogensis **2**
62. The mitotic spindle and contractile ring are **perpendicular** to each other.
63. Gastrulation in Snails is accomplished primarily by **epiboly.**
64. The first cell movement of fish gastrulation is the **epiboly** of the blastoderm.
65. This fetal organ the **Chorion** is derived primarily from embryonic trophoblast cells.
66. Secondary neurulation involves the making of a **medullary cord** and its subsequent hollowing into neural tube
67. The mantle zone cells differentiate into both **neurons** and **glia.**
68. The cerebral cortex in humans has **six** layers.
69. Paracrine factors such as **TGF** and FGF7 are important in normal skin development.
70. The **fifth** phase involves the invasion of the cartilage model by blood vessels.

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39. D
40. Fertilization in the absence of sex occur through the __Binary Fission__
41. The macromeres divide meridionally, forming a tier of eight (8) cells
42. cells, derived from the micromeres, are called the primary mesenchyme.
43. Molluscus follow which type of cleavage ____spiral holoblastic cleavage
44. A mammalian blastula is called a blastocyst.
45. Which of the following undergo discoidal cleavage __fishes, reptiles and birds
46. The egg jelly factors that initiate the Acrosomal reaction in sea urchins are often highly specific to each species
47. The Metencephalon gives rise to the cerebellum
48. In primary neurulation, the original ectoderm is divided into three sets of cells
49. The growing region at the tip of the lip is called the chordoneural hinge
50. Neural Tube will form ___Brain and spinal cord
51. Each ICM (Inner Cell Mass) cell has the same potency
52. Variable cleavages produce no invariant fate assignments to cells
53. In early embryos of these insects, cell division is not complete
54. The neural crest cells originate at the dorsal most region of the neural tube
55. The human β-globin gene has an enhancer in its 3′ UTR, roughly 700 base pairs
56. the SOX9 gene, which encodes a DNAbinding protein
57. the entire region of calcified spicules becomes surrounded by compact mesenchymal cells that form the Periosteum
58. The second step is the alignment of the myoblasts together into chains
59. Cleavages sphere is known as __Blastula
60. Osteoclasts are probably derived from the same precursors as macrophage blood cells
61. charby desired to know the cause of birth defects
62. The direct conversion of mesenchymal tissue into bone is called intramembranous ossification
63. Those cells that remain integral components of the neural tube lining become ependymal cells
64. The original type of trophoblast cells constitute a layer called the cytotrophoblast
65. The major structural characteristic of avian, reptilian, & mammalian gastrulation is the primitive streak
66. The Blastomere cell cycle is governed by the synthesis and degradation of_____cyclin
67. Both have Anurans __and _. __Toad and frog
68. They have a notochord as larvae (chordates), but they lack vertebrae.
69. Acrosomal vesicles is derived from Golgi apparatus and contains __Enzymes__
70. The large subunit of MPF is __cyclin B
71. ZP3 is the specific glycoprotein in the mouse
72. MITF necessary for the production of __Pigment cells
73. Mutation of sox9 gene cause ____camptomelic dysplasia
74. Between the blastoderm and the yolk is a space called the sub-germinal cavity
75. The second step of gastrulation involves the involution of the mesoderm
76. The first cleavage furrow appears centrally in the blastodisc
77. Fertilization occurs in the ampulla of the oviduct
78. Mother provides fetus with __nutrients and oxygen
79. The first segregation of cells within the inner cell mass results in the formation of the hypoblast (sometimes called the primitive endoderm)
80. The fetal portion of placenta formed by __Chorion__
81. Paracrine factor instruct myotome cells to become__ Muscles
82. When myoblasts stop dividing, secrete fibronectin

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83. Retinoic Acid comes from **Somites**
84. Hox genes pattern the anterior-posterior axis and help to specify positions along that **axis**
85. The telencephalon will eventually form the cerebral hemispheres, and the diencephalon will form the thalamic and **hypothalamic** brain regions
86. Endochondral ossification involves the formation of **cartilage**
87. J.Hammerling exchanged nuclei b/w two **morphologically** distinct species
88. When plus individuals initiate fusion by extending a fertilization tube. Which contacts & fuses with a specific site on the **minus** individual.
89. Promoter is **enhancer** which is thought to regulate transcription.
90. Identical twins share a common **Chorion**
91. In Birds, **pH** differences appear crucial for distinguishing dorsal from ventral
92. Which regulate function of Promoter Enhancer
93. The space between two neurons, where transmitter are released **synapse**
94. integral components of the neural tube lining become **ependymal**
95. Mantle cells differentiate into **Gray matter**
96. The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the area **opaca.**
97. During fertilization, the **actin** molecules fuse to form the long microfilaments.
98. A condition in which cell lack the instructions for development is **naive**
1. metamorphosis is initiated by hormones from the tadpole's **thyroid gland**
2. Brain has **100** billion neurons
3. Release of Ca from intracellular storage controlled by dye **Aequorin**
4. Nervous system develop by **Neural Tube**
5. Allows the mRNA to exit the nucleus **poly a tail**
6. Contact and ___between sperm and egg **Recognition**
7. The first evidence suggesting the importance of sperm in reproduction came from a series of experiments performed by ____in the late 1700s. **Lazzaro Spallanzani**
8. The early embryonic cells need a supply of energy and amino acids which is provided by **yolk** proteins in the egg.
9. The depression is called the primitive **groove**

ZOO501 CURRENT MCQS MIDTERM (2018) UPDATED

1. Red blood cells make Globins
2. how many strategies for developmental commitment 2
3. Ribosomes produce in sea urchin 10^{12}
4. Resact is isolated from egg jelly of sea urchin
5. ___is a discipline that studies embryonic & other developmental processes Developmental Biology
6. 0nly small %age of genome expressed in particular cell
7. Digestive & respiratory system formed by Endoderm
8. Progesterone work for signals the egg to resume its meiotic division
9. Insect characteristics are ..... specification syncytial
10. In Chlamydomonas haploid chromosome has.....copy one
11. Modes of commitments are..... 3
12. After fertilization developmental process proceed by Cleavage
13. Ca ion responsible for cortical granules stores in Endoplasmic Reticulum of Eggs
14. Seawater has a particularly high ___ ion concentration Na (Sodium)
15. There are ___approaches of anatomy 4(four)
16. The major cell adhesion molecules appear to be the Cadherins.
17. It is estimate ____egg is 25000 to 50000 mRNA. *Sea Urchin*
18. Fusion of ____and sperm cell plasma membrane. Egg
19. Resact diffuse readily in sea water.
20. The requirement of capacitance varies from species to Species.

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